LECTURE SERIES · DATABASE SYSTEMS

# Emerging Topics & Analytics

*Database Systems — Modern Data Platforms & Analytics*

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Crypto · NoSQL · Cloud · Replicas · Snapshots · Warehousing · Analytics · Governance

TODAY'S ROADMAP

# Emerging Topics & Analytics

Nine stops on the tour — from protecting bytes with cryptography to obeying the laws that govern them.

## Cryptography

Encryption, hashing & digital signatures secure data at rest and in transit.

## NoSQL & Distributed DBs

Key-value, document & graph stores scale out across many nodes.

## Cloud Databases

Managed services (RDS, Cosmos, BigQuery) — pay-as-you-go, auto-scaling.

## Read Replica

Copies of the primary serve read traffic — more throughput, less load.

## Database Snapshot

A point-in-time, read-only copy — instant restore & safe reporting.

## Data Warehousing

Integrated historical data organized in star schemas for OLAP queries.

## Data Analytics

From dashboards to ML — turning stored data into decisions.

## Data Governance

Ownership, quality & catalogs — rules for how data is managed.

## Privacy & Compliance

GDPR-style laws: consent, minimization & the right to be forgotten.

SECRETS THAT STAY SECRET

# Cryptography

Cryptography turns readable data into protected data — the mathematical backbone of database security.

crypto.txt

```
-- symmetric encryption (shared key)
AES-256("Rakib", key) → 9f2a...c41d

-- one-way hashing (no key, no undo)
SHA-256("password")   → 5e88...1542

-- asymmetric: public / private key pair
sign(doc, private)  verify(doc, public)
```

## Encryption

Protects data at rest (files, backups) and in transit (TLS) — stolen bytes stay unreadable.

## Hashing

Stores passwords & verifies integrity — one-way, so the original can never be recovered.

## Digital Signatures

Prove who wrote the data and that nobody changed it — authenticity + integrity.

WHEN ONE SERVER IS NOT ENOUGH

# NoSQL & Distributed Databases

NoSQL systems trade rigid schemas for flexibility and horizontal scale across many machines.

## Key-Value

A giant hash map: get / put by key. Blazing fast caching. (Redis, DynamoDB)

## Document

JSON-like documents, flexible schema per record. (MongoDB, CouchDB)

## Column-Family

Wide rows stored by column for huge write loads. (Cassandra, HBase)

## Graph

Nodes & edges for relationship-heavy data. (Neo4j)

**CAP theorem:** a distributed store can guarantee only two of Consistency, Availability and Partition tolerance — every system picks its trade-off.

DATABASES AS A SERVICE

# Cloud Databases

Instead of buying servers, you rent a database — the provider handles hardware, patching, backups and scaling.

## Managed Relational

Familiar SQL engines run for you — provisioning, patching & backups automated.

`e.g.  Amazon RDS, Azure SQL`

## Cloud-Native NoSQL

Globally distributed, multi-model stores with SLA-backed latency.

`e.g.  DynamoDB, Cosmos DB`

## Serverless Analytics

Query petabytes with no infrastructure — pay per query scanned.

`e.g.  BigQuery, Snowflake`

**Why it wins:** elastic scaling, pay-as-you-go pricing, built-in high availability — but watch cost, vendor lock-in and data residency.

SCALING THE READS

# Read Replica

One primary handles all writes; copies of it serve the read traffic — more throughput without touching the app logic.




## Scale reads

Dashboards, reports & APIs read from replicas — the primary stays fast for writes.

## Replication lag

Copies trail the primary by moments — reads can be slightly stale (eventual consistency).

## Failover

If the primary dies, a replica is promoted — downtime shrinks to seconds.

FREEZING A MOMENT

# Database Snapshot

A snapshot is a read-only, point-in-time copy of the database — created in seconds, not hours.

## Copy-on-Write magic

The snapshot starts empty; only when a page changes is its old version copied out — so creation is instant and storage stays small.

## How it is used

Report against a frozen, consistent view; test risky migrations; restore instantly to the moment before a mistake.

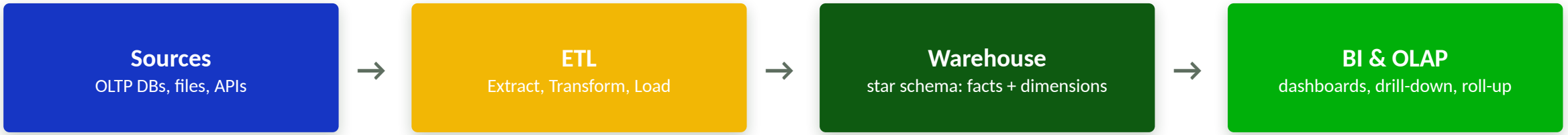
snapshot.sql

```sql
-- SQL Server example
CREATE DATABASE Uni_Snap ON (NAME = Uni, FILENAME = 'uni.ss')
    AS SNAPSHOT OF University;
```

A HOME FOR HISTORY

# Data Warehousing

A warehouse integrates years of data from many sources and shapes it for analysis, not transactions.




**OLTP vs OLAP:** operational databases optimize many small writes; the warehouse optimizes huge read-only analytical queries.

**Star schema:** a central fact table (sales) surrounded by dimension tables (date, product, region) — simple joins, fast aggregation.

FROM DATA TO DECISIONS

# Data Analytics

Analytics climbs a ladder of questions — each step adds more value and more sophistication.

1

### Descriptive

What happened? Reports & dashboards.

2

### Diagnostic

Why did it happen? Drill-down & correlation.

3

### Predictive

What will happen? ML & forecasting.

4

### Prescriptive

What should we do? Optimization.

**The pipeline:** collect → clean → store (warehouse / lake) → analyze → visualize — SQL remains the analyst's first language.

RULES FOR THE DATA ESTATE

# Data Governance

Governance answers: who owns this data, how good is it, where is it, and how long do we keep it?

## Ownership & Stewardship

Every dataset has a named owner accountable for its accuracy and access.

## Data Quality

Completeness, accuracy, timeliness — measured, monitored and fixed at the source.

## Catalog & Metadata

A searchable inventory: what data exists, what it means, where it came from (lineage).

## Lifecycle & Retention

Classify, archive and delete on schedule — keeping data forever is a liability.

**Why bother:** trusted data → trusted analytics. Governance is what makes "data-driven" more than a slogan.

THE LAW MEETS THE DATABASE

# Privacy & Compliance

Privacy laws such as GDPR turn good data handling from a courtesy into a legal obligation.

## Consent & Purpose

Collect data only with permission, and use it only for the stated purpose.

## Data Minimization

Store the least data needed, for the shortest time — less data, less risk.

## Right to be Forgotten

Users can demand erasure — your schema must make true deletion possible.

## Breach Notification

Detect, report (within 72h under GDPR) and document every data breach.

**The landscape:** GDPR (EU), CCPA (California), Bangladesh's Data Protection framework — design for compliance from day one, not as a patch.

KEY TAKEAWAYS

# Recap

### Cryptography

Encrypt, hash & sign to protect data

### NoSQL & Distributed

Flexible models that scale out (CAP)

### Cloud, Replicas & Snapshots

Managed DBs, scaled reads, instant copies

### Data Warehousing

Sources → ETL → star schema → OLAP

### Data Analytics

Descriptive → predictive → prescriptive

### Governance & Privacy

Ownership, quality & GDPR compliance

# Thank You

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